Prepared for Hsin Chuan Semiconductor Academy AI Agent Practical Competition, 2026.

Circuit.AI Hardware Splicer Architecture

Core Claim

Circuit.AI is an evidence-gated AI agent for circuit board understanding, repair, reuse, and hardware recomposition. It does not treat visual model output as electrical truth. It converts model output into measurement plans and only promotes a board to repair authority after measured evidence and bench outcomes.

Architecture

```
flowchart LR
  A[Multi-photo board evidence] --> B[Visual evidence extraction]
  C[Public datasheets and pinouts] --> D[Reference topology]
  B --> E[Board evidence v1]
  D --> E
  E --> F[Visual topology hypothesis]
  F --> G[Bench measurement packet]
  G --> H[Measured topology evidence]
  H --> I[Simulation and safety envelope]
  I --> J[Controlled bench outcome]
  J --> K[Production release manifest]
  K --> L[Production authority casefile]

  M[Qwen or vision model] --> B
  N[DeepSeek/Copilot reasoner] --> F
  O[Deterministic verifier] --> L
```

Evidence Authority Levels

Level	Meaning	Allowed Output
`visual_candidate`	photos and reference hints only	capture plan and measurement queue
`measured_topology`	trusted pinout/rail measurements exist	scoped topology and simulation checks

Level	Meaning	Allowed Output
`electrical_simulation`	measured envelope passes deterministic checks	bench protocol and guarded bring-up
`controlled_bench`	first-power/function proof recorded	controlled reuse claim
`production_repair`	release manifest and artifacts complete	scoped production repair/reuse authority

---

Current Live Demo

Frontend:

```
/showcase?state=release
```

Backend:

```
POST /hardware/production-casefile/run
```

The showcase demonstrates:

- evidence packet selection
- board topology visualization
- measurement closure
- authority ladder
- live production casefile response
- runtime model/budget safety strip

Model Strategy

Models help interpret and plan:

- Qwen VL: candidate board evidence from multi-photo input.
- DeepSeek/Copilot-style reasoners: structured advisory reasoning.
- Local deterministic verifier: final authority gate.

The system is designed so model output can accelerate work but cannot authorize unsafe hardware actions alone.

Competition Deliverable

The final demo will show a CH340C USB-serial board moving through:

```
reference-only evidence -> blocked authority
measured topology -> simulation-ready authority
release evidence -> production repair authorized
```